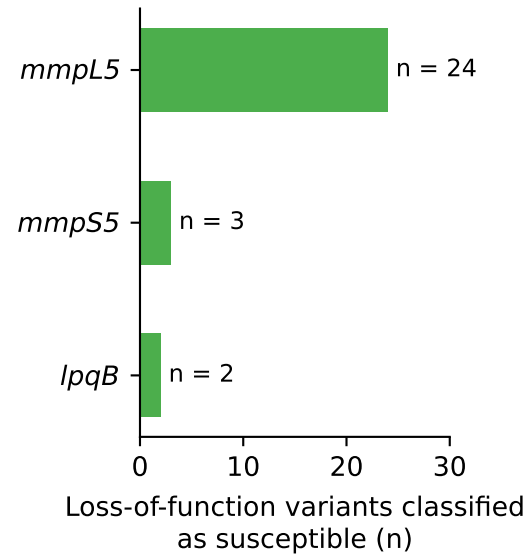
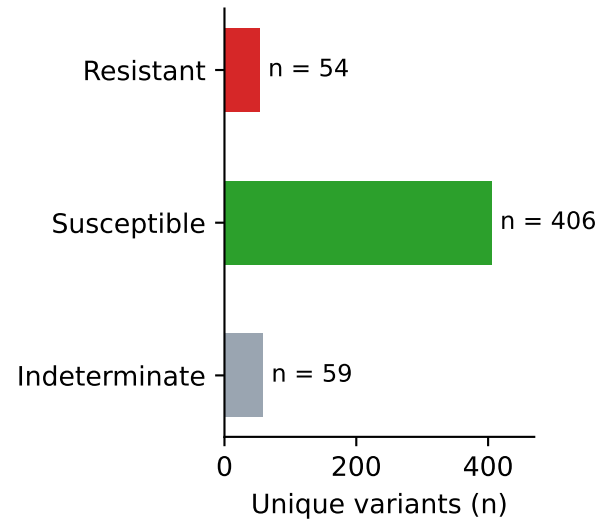


C. LoF in structural-integrity genes → susceptible



D. Variant-level pipeline outcome



*mtrA/mtrB*: same sensitizing logic. *glpK/Rv1979c* (exploratory): recorded without a resistance call = 43 full-pipeline calls + 11 screening-stage frameshifts (9 *Rv0678*, 2 *pepQ*).